ipv4: validate checksum_start before completing checksum

ipv4: validate checksum_start before completing checksum - Michael S. Tsirkin, Linux kernel project.

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Content

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```
From 6785011f8b16abf40b1e9d8b2e332232b3e68822 Mon Sep 17 00:00:00 2001
From: "Michael S. Tsirkin" <mst@redhat.com>
Date: Wed, 7 Oct 2026 18:42:49 -0400
Subject: [PATCH] ipv4: validate checksum_start before completing checksum

If a packet with bad checksum metadata gets into the ipv4 stack,
skb_checksum_help can corrupt the network header and cause a bunch of
mischief.

This was discovered and reported by Paulos, and has been reporoduced
by others independently since.

We really shouldn't allow such packets in, but as a defence
in depth measure, let's also check before we complete the checksum.

A more complete validation at input is forthcoming, but needs more work.

Cc: stable@vger.kernel.org
Fixes: f43798c27684 ("tun: Allow GSO using virtio_net_hdr")
Fixes: bfd5f4a3d605 ("packet: Add GSO/csum offload support.")
Reported-by: Paulos Yibelo <habte.yibelo@gmail.com>
Closes: https://lore.kernel.org/netdev/20260922030310.8684-2-habte.yibelo@gmail.com/
Signed-off-by: Michael S. Tsirkin <mst@redhat.com>
Reviewed-by: Willem de Bruijn <willemb@google.com>
Link:
https://patch.msgid.link/0a012b4923e189c4c593ef4f471e5ff0edbe9030.1791412497.git.mst@redhat.com
Signed-off-by: Jakub Kicinski <kuba@kernel.org>
---
 include/net/ip.h                           |  7 +++++++
 net/bridge/netfilter/nf_conntrack_bridge.c | 10 +++++++---
 net/ipv4/ip_output.c                       | 10 +++++++---
 net/netfilter/nfnetlink_queue.c            |  7 +++++++
 net/netfilter/xt_CHECKSUM.c                |  6 +++++-
 net/xfrm/xfrm_output.c                     | 16 ++++++++++++----
 6 files changed, 44 insertions(+), 12 deletions(-)

diff --git a/include/net/ip.h b/include/net/ip.h
index 6f602df72ee621..d07c2c5730240b 100644
--- a/include/net/ip.h
+++ b/include/net/ip.h
@@ -74,6 +74,13 @@ static inline unsigned int ip_hdrlen(const struct sk_buff *skb)
         return ip_hdr(skb)->ihl * 4;
 }
```

```
+static inline int ip_check_csum_start(const struct sk_buff *skb)
+{
+         if (unlikely(skb->csum_start < skb->network_header + ip_hdrlen(skb)))
+                  return -EINVAL;
+         return 0;
+}
+
 struct ipcm_cookie {
          struct sockcm_cookie         sockc;
          __be32                       addr;
diff --git a/net/bridge/netfilter/nf_conntrack_bridge.c
b/net/bridge/netfilter/nf_conntrack_bridge.c
index 7ecb8a26bfa3a7..b5444335b86f2b 100644
--- a/net/bridge/netfilter/nf_conntrack_bridge.c
+++ b/net/bridge/netfilter/nf_conntrack_bridge.c
@@ -39,9 +39,13 @@ static int nf_br_ip_fragment(struct net *net, struct sock *sk,
          int err = 0;

          /* for offloaded checksums cleanup checksum before fragmentation */
-         if (skb->ip_summed == CHECKSUM_PARTIAL &&
-             (err = skb_checksum_help(skb)))
-                  goto blackhole;
+         if (skb->ip_summed == CHECKSUM_PARTIAL) {
+                  err = ip_check_csum_start(skb);
+                  if (!err)
+                           err = skb_checksum_help(skb);
+                  if (err)
+                           goto blackhole;
+         }

          iph = ip_hdr(skb);

diff --git a/net/ipv4/ip_output.c b/net/ipv4/ip_output.c
index a24cc8ee11d3ea..eaa6fabda347e9 100644
--- a/net/ipv4/ip_output.c
+++ b/net/ipv4/ip_output.c
@@ -771,9 +771,13 @@ int ip_do_fragment(struct net *net, struct sock *sk, struct sk_buff
*skb,
          int err = 0;

          /* for offloaded checksums cleanup checksum before fragmentation */
-         if (skb->ip_summed == CHECKSUM_PARTIAL &&
-             (err = skb_checksum_help(skb)))
```

```
-                   goto fail;
+         if (skb->ip_summed == CHECKSUM_PARTIAL) {
+                   err = ip_check_csum_start(skb);
+                   if (!err)
+                             err = skb_checksum_help(skb);
+                   if (err)
+                             goto fail;
+         }

          /*
           *        Point into the IP datagram header.
diff --git a/net/netfilter/nfnetlink_queue.c b/net/netfilter/nfnetlink_queue.c
index a3bc0028005109..477fc632657f4c 100644
--- a/net/netfilter/nfnetlink_queue.c
+++ b/net/netfilter/nfnetlink_queue.c
@@ -40,6 +40,7 @@
 #include <linux/udp.h>
 #include <net/gre.h>
 #include <net/gso.h>
+#include <net/ip.h>
 #include <net/sock.h>
 #include <net/tcp_states.h>
 #include <net/netfilter/nf_queue.h>
@@ -674,6 +675,12 @@ static int nfqnl_put_bridge(struct nf_queue_entry *entry, struct
sk_buff *skb)

 static int nf_queue_checksum_help(struct sk_buff *entskb)
 {
+         if (entskb->protocol == htons(ETH_P_IP)) {
+                   if (!pskb_network_may_pull(entskb, sizeof(struct iphdr)) ||
+                       ip_check_csum_start(entskb))
+                             return -EINVAL;
+         }
+
          if (skb_csum_is_sctp(entskb))
                    return skb_crc32c_csum_help(entskb);

diff --git a/net/netfilter/xt_CHECKSUM.c b/net/netfilter/xt_CHECKSUM.c
index 9d99f5a3d1764b..1fb0f8118404d6 100644
--- a/net/netfilter/xt_CHECKSUM.c
+++ b/net/netfilter/xt_CHECKSUM.c
@@ -15,6 +15,7 @@

 #include <linux/netfilter_ipv4/ip_tables.h>
```

```
 #include <linux/netfilter_ipv6/ip6_tables.h>
+#include <net/ip.h>

 MODULE_LICENSE("GPL");
 MODULE_AUTHOR("Michael S. Tsirkin <mst@redhat.com>");
@@ -25,8 +26,11 @@ MODULE_ALIAS("ip6t_CHECKSUM");
 static unsigned int
 checksum_tg(struct sk_buff *skb, const struct xt_action_param *par)
 {
-        if (skb->ip_summed == CHECKSUM_PARTIAL && !skb_is_gso(skb))
+        if (skb->ip_summed == CHECKSUM_PARTIAL && !skb_is_gso(skb)) {
+                if (xt_family(par) == NFPROTO_IPV4 && ip_check_csum_start(skb))
+                        return NF_DROP;
                 skb_checksum_help(skb);
+        }

         return XT_CONTINUE;
 }
diff --git a/net/xfrm/xfrm_output.c b/net/xfrm/xfrm_output.c
index e305ba32e356b7..f1f612cd7b438d 100644
--- a/net/xfrm/xfrm_output.c
+++ b/net/xfrm/xfrm_output.c
@@ -823,16 +823,22 @@ int xfrm_output(struct sock *sk, struct sk_buff *skb)
         }

         if (skb->ip_summed == CHECKSUM_PARTIAL) {
-                err = skb_checksum_help(skb);
-                if (err) {
-                        XFRM_INC_STATS(net, LINUX_MIB_XFRMOUTERROR);
-                        kfree_skb(skb);
-                        return err;
+                if (skb->protocol == htons(ETH_P_IP)) {
+                        err = ip_check_csum_start(skb);
+                        if (err)
+                                goto error;
                 }
+                err = skb_checksum_help(skb);
+                if (err)
+                        goto error;
         }

 out:
         return xfrm_output2(net, sk, skb);
+error:
```

```
+         XFRM_INC_STATS(net, LINUX_MIB_XFRMOUTERROR);
+         kfree_skb(skb);
+         return err;
 }
 EXPORT_SYMBOL_GPL(xfrm_output);
```

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